

BUS 713 Financial Analytics

Assessment 2

Predictive Modelling and Portfolio Evaluation

1. You are a financial analyst advising a portfolio manager on the risk and expected return profile of two stocks of your choice. Using R, retrieve historical monthly stock price data and estimate each stock's beta relative to the Nasdaq index. Apply the Capital Asset Pricing Model (CAPM) to calculate the expected return for each stock, and use the Security Market Line (SML) to assess whether each stock appears overvalued or undervalued relative to its systematic risk. Discuss the empirical limitations of the CAPM and reflect on what these mean for how much confidence an investor should place in your findings.

Next, compute the correlation between the monthly returns of your two chosen stocks and discuss what this implies for holding them together in a portfolio. Does combining these two stocks offer meaningful diversification benefits? How does this inform your recommendation to the portfolio manager?

In the second part of your answer, use your monthly price data to forecast the closing price of one of your chosen stocks over the next six months. You are free to choose any forecasting approach you consider appropriate. Justify your choice of method, plot your forecast against the historical data, and evaluate how reliable your forecast is. Reflect on what your forecast implies for a portfolio manager making decisions about this stock.

In addition, briefly explain how either a futures contract or an options contract could be used to hedge the downside risk of holding one of your chosen stocks. Reflect on the practical advantages and limitations of the hedging strategy you describe.

2. Your firm is evaluating a five-year investment project requiring an upfront capital outlay of \$500,000, with expected annual revenues of \$200,000 and annual costs of \$100,000. Using R, compute the NPV and IRR under a base case assumption and interpret both measures clearly.

If your cash flows are locked in through long-term contracts, your base case may be sufficient to support a decision. However, if cash flows are uncertain, a single NPV figure may give management a false sense of precision. Extend analysis to reflect this uncertainty and provide management with a fuller picture of possible outcomes. You can consider any scenario; however, you must justify your choice.

Submission Components

1. Video Presentation

You must submit a 6–10-minute recorded video presentation aimed at management and other stakeholders. Your presentation should address both questions.

Your presentation should include:

- Your approach
- Explanations as to why you chose this approach
- How your code connects to the solution and conclusions
- Summary of key findings
- Recommendations

You must:

- be visible on camera throughout
- show your student ID at the start

2. R script

It should be well structured and reproducible. Comments should explain the key steps.

3. Executive Summary

An executive summary of your work, the challenges you faced in completing this exercise, and how you might apply the tools and techniques you used in this exercise in your work and life.

The summary should briefly answer the following questions:

- What did you analyse?
- What did you find?
- What would you recommend?
- What are the limitations?

Your executive summary should be approximately 250 words.